

Exercise #7

Introduction To Artificial Intelligence

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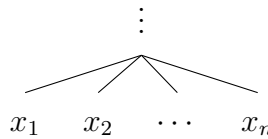
Due Date: March 9th, 2018

1 x To $ax + b$ Leaf Node Transformation

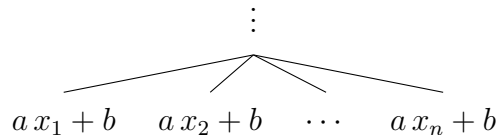
Problem Statement. *Prove that with a positive linear transformation of leaf values (i.e., transforming a value x to $ax + b$ where $a > 0$), the choice of move remains unchanged in a game tree, even when there are chance nodes.*

Proof. We will prove the following via proof of mathematical induction.

Base Case We will take the base case with a lead nodes, so the transformation would be as follows:



would transform into:



Recalling the definition of MINIMAX, we get the following:

$$\text{MINIMAX}(s) = \begin{cases} \text{MAX'S UTILITY}(s) & \leftrightarrow \text{TERMINAL-TEST}(s) \\ \max_{a \in \text{ACTIONS}(s)} \text{MINIMAX}(\text{RESULT}(s, a)) & \leftrightarrow \text{PLAYER}(s) = \text{MAX} \\ \min_{a \in \text{ACTIONS}(s)} \text{MINIMAX}(\text{RESULT}(s, a)) & \leftrightarrow \text{PLAYER}(s) = \text{MIN} \end{cases} \quad (1)$$

Recalling that ACTIONS returns a set of values (let us name them $x_1, x_2, \dots, x_n$) our min() and max() values can be rewritten as:

$$\begin{aligned} \min(x_1, x_2, \dots, x_n) &= x_p \\ \max(x_1, x_2, \dots, x_n) &= x_p \end{aligned}$$

where x_p is the chosen node. For the probability, we get

$$p(x_1) + p(x_2) + \cdots + p(x_n)$$

Applying the transformation, we get the following:

$$\begin{aligned}\min(a x_1 + b, a x_2 + b, \dots, a x_n + b) &= a x_p + b \\ \max(a x_1 + b, a x_2 + b, \dots, a x_n + b) &= a x_p + b \\ p_1(a x_1 + b) + p_2(a x_2 + b) + \cdots + p_n(a x_n + b)\end{aligned}$$

From this, we factor out a and b :

$$\begin{aligned}a \min(x_1, x_2, \dots, x_n) + b &= a x_p + b \\ a \max(x_1, x_2, \dots, x_n) + b &= a x_p + b \\ a(p_1 x_1 + p_2 x_2 + \cdots + p_n x_n) + b\end{aligned}$$

Upon which we can clearly see a ($\forall a \in \mathbb{R}^+$) and b have no choice on which node gets chosen.

Inductive Step As we scale our problem up for MINIMAX (Equation ??), we see that if the decision of the terminal node is unchanged, the decision at the root will be changed. The same terminal value will be returned; and as we go up the game tree, each ply acts as a terminal.

□